Machine Learning and Data Mining (ST443)

Zoltán Szabó

LSE

Content: linear regression (Ch 3 @ ISLP, Ch 3.2 @ ESL)

- probability recap,
- objective, solution, projection interpretation,
- testing, confidence interval,
- measure of fit,
- linear regression tweaks, towards non-linear methods.

Probability recap: continuous distributions on $\mathbb{R}^d$

Representation of $\mathbb{P}$: pdf (probability density function) f

1. non-negative:

$$f(\mathbf{x}) \geq 0 \text{ for all } \mathbf{x} \in \mathbb{R}^d.$$

2. normalized:

$$\int_{\mathbb{R}^d} f(\mathbf{x}) \mathrm{d}\mathbf{x} = 1.$$

Probability recap: continuous distributions on $\mathbb{R}^d$

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Meaning: $X \sim \mathbb{P}$, with pdf f_X, for 'any' set $B \subseteq \mathbb{R}^d$

$$\mathbb{P}(X \in B) = \int_B f_X(\mathbf{x}) \mathrm{d}\mathbf{x}.$$

Continuous distribution – continued

Alternative representations:

- Cdf (cumulatived density function): $X \sim \mathbb{P}$,

$$F(\mathbf{z}) = \mathbb{P}(X < \mathbf{z}) = \mathbb{P}(X \in \underbrace{\times_{j=1}^{d}(-\infty, z_j)}_{=B\text{: interval}}), \quad \mathbf{z} = (z_j)_{j=1}^{d} \in \mathbb{R}^d.$$

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- Characteristic function ($i = \sqrt{-1}$):

$$\phi_X(\mathbf{t}) = \mathbb{E}\left[e^{i\langle \mathbf{t}, X\rangle}\right], \quad \mathbf{t} \in \mathbb{R}^d.$$

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At the moment

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Usage of pdf: $d = 1$

- Expectation, moments:

$$\mathbb{E}[X] = \int_{\mathbb{R}} x f(x) \mathrm{d}x, \quad \mathbb{E}[X^n] = \int_{\mathbb{R}} x^n f(x) \mathrm{d}x \quad (n \in \mathbb{Z}^+).$$

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- Covariance of $(X, Y) \in \mathbb{R} \times \mathbb{R}$:

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$$\mathrm{var}(X) = \mathrm{cov}(X, X), \qquad \sigma(X) = \sqrt{\mathrm{var}(X)}.$$

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- Correlation:

$$\mathrm{corr}(X, Y) = \frac{\mathrm{cov}(X, Y)}{\sigma(X)\sigma(Y)}.$$

Properties ($d = 1$)

- Covariance, with constants $a, b \in \mathbb{R}$:

$$\begin{aligned} \operatorname{cov}(X, a) &= 0, \\ \operatorname{cov}(X, Y) &= \operatorname{cov}(Y, X), \\ \operatorname{cov}(aX, bY) &= ab \operatorname{cov}(X, Y), \\ \operatorname{cov}(X + a, Y + b) &= \operatorname{cov}(X, Y). \end{aligned}$$

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- Correlation (Pearson's correlation coefficient), with constant $a, b \in \mathbb{R}$:

$$\operatorname{corr}(X, Y) \in [-1, 1], \quad |\operatorname{corr}(X, Y)| = 1 \Leftrightarrow X = aY + b.$$

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$$\mathrm{var}(X) = \mathrm{cov}(X, X).$$

Properties ($d \geq 1$)

- Mean ($\mathbf{A} \in \mathbb{R}^{d_1 \times d_2}$, $\mathbf{b} \in \mathbb{R}^{d_2}$, $X \in \mathbb{R}^{d_2}$):

$$\mathbb{E}[\mathbf{A}X + \mathbf{b}] = \mathbf{A}\mathbb{E}[X] + \mathbf{b} \text{ (under affine transformation)}.$$

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Celebrities in the area: normal distribution ($d \geq 1$)

- $\mathbb{R}^d \ni X \sim \mathcal{N}(\mathbf{m}, \mathbf{\Sigma})$ has pdf

$$f_X(\mathbf{x}) = (2\pi)^{-d/2}[\det(\mathbf{\Sigma})]^{-1/2} e^{-\frac{1}{2}(\mathbf{x}-\mathbf{m})^\top \mathbf{\Sigma}^{-1}(\mathbf{x}-\mathbf{m})}.$$

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- Under affine transformation:

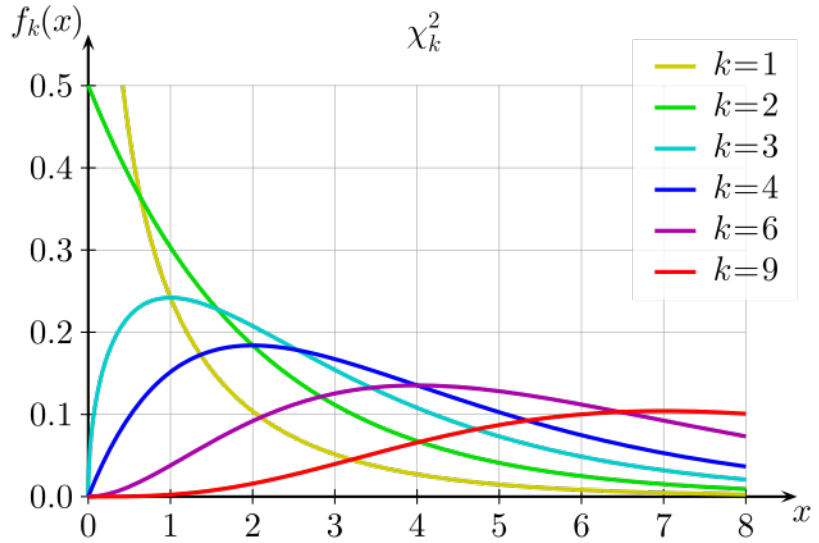
$$\mathbf{A}X + \mathbf{b} \sim \mathcal{N}\left(\mathbf{Am} + \mathbf{b}, \mathbf{A\Sigma A}^\top\right).$$

Celebrities+: χ^2 distribution ($d = 1$)

- Definition ($k \in \mathbb{Z}^+$ degrees of freedom):

$$\chi^2_k = \sum_{i=1}^{k} X_i^2,$$

where $X_1, \ldots, X_k$ are independent and $\mathcal{N}(0, 1)$-distributed.

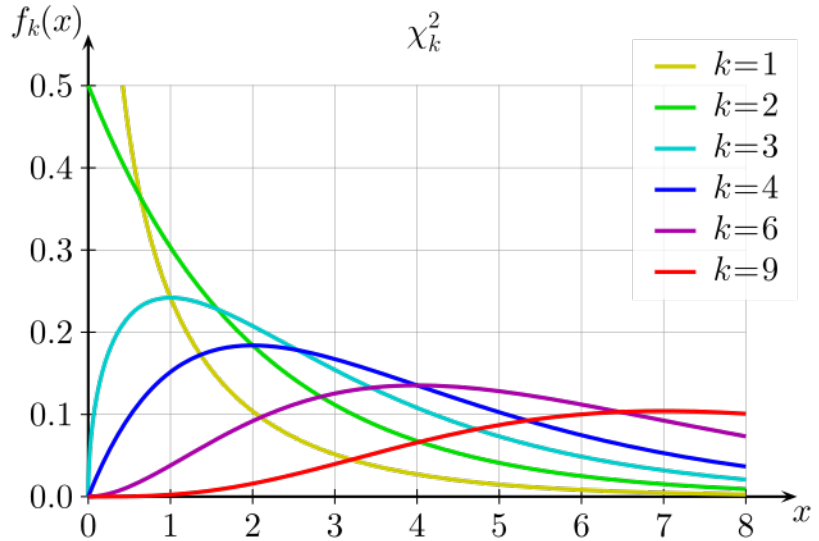


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- Pdf:

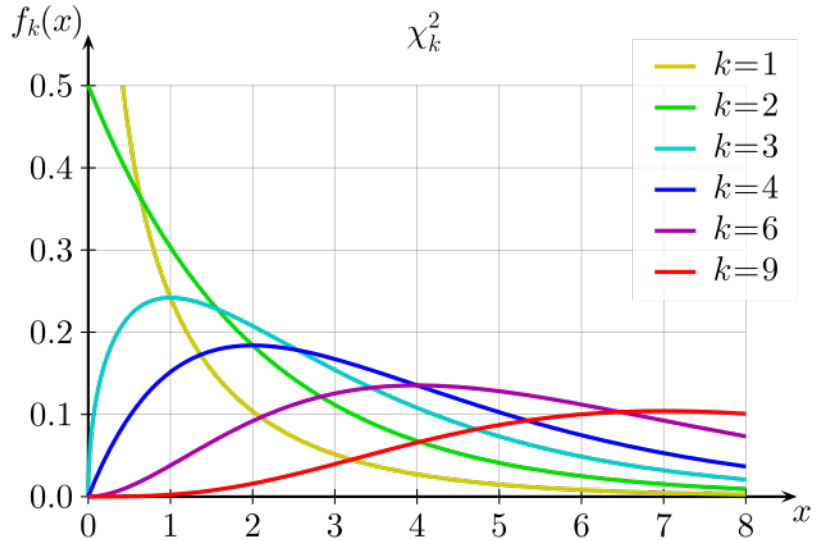
$$f_k(x) = \frac{1}{2^{\frac{k}{2}}\Gamma\left(\frac{k}{2}\right)} x^{\frac{k}{2}-1} e^{-\frac{x}{2}},\ x \in (0, \infty), \quad \Gamma(z) = \int_{\mathbb{R}_{\geq 0}} t^{z-1} e^{-t} \mathrm{d}t\ (z > 0).$$

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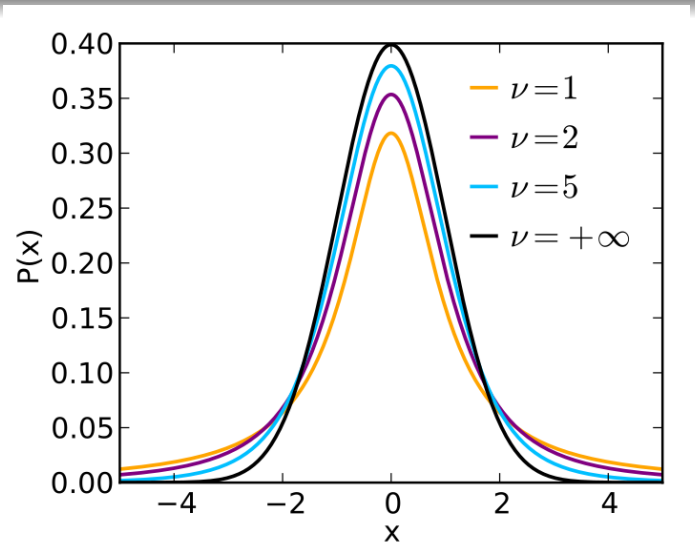


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- Mean, variance: $\mathbb{E}[X] = k$, $\mathrm{var}[X] = 2k$.

Celebrities+: t-distribution ($d = 1$)

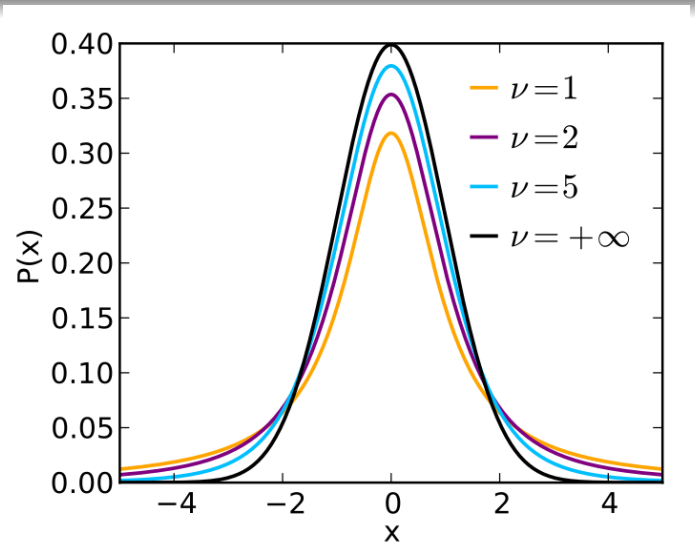


- Pdf (ν = degrees of freedom > 0):

$$f_\nu(x) = \frac{1}{\sqrt{\nu}B\left(\frac{1}{2},\frac{\nu}{2}\right)} \frac{1}{\left(1+\frac{x^2}{\nu}\right)^{\frac{\nu+1}{2}}} \quad (x \in \mathbb{R}),$$

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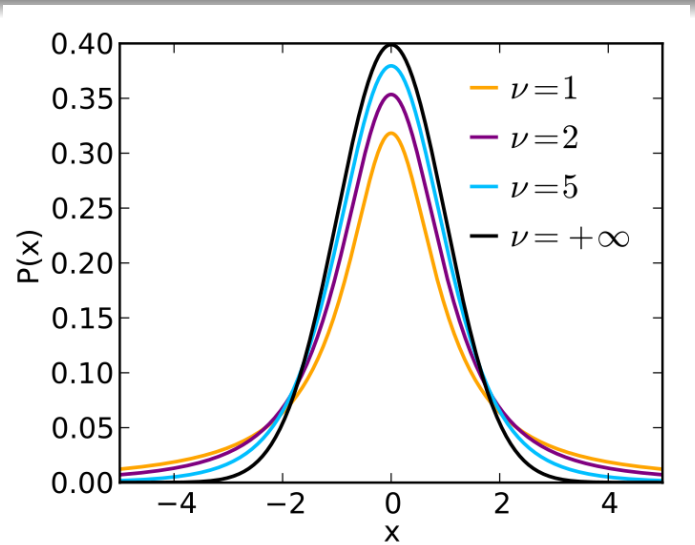
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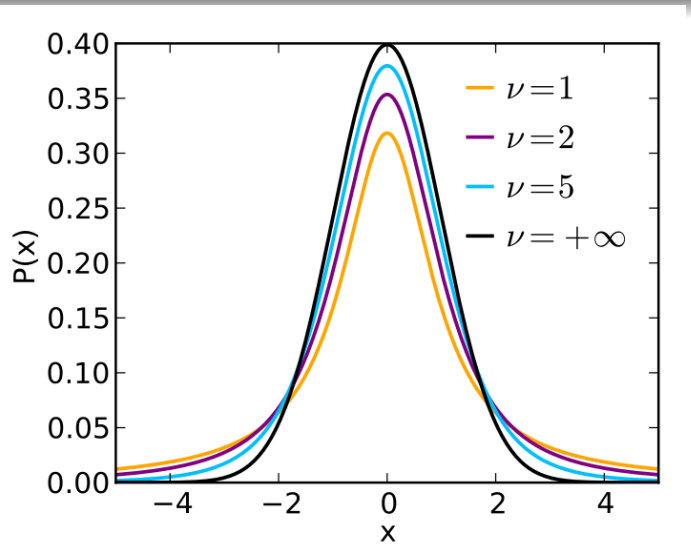
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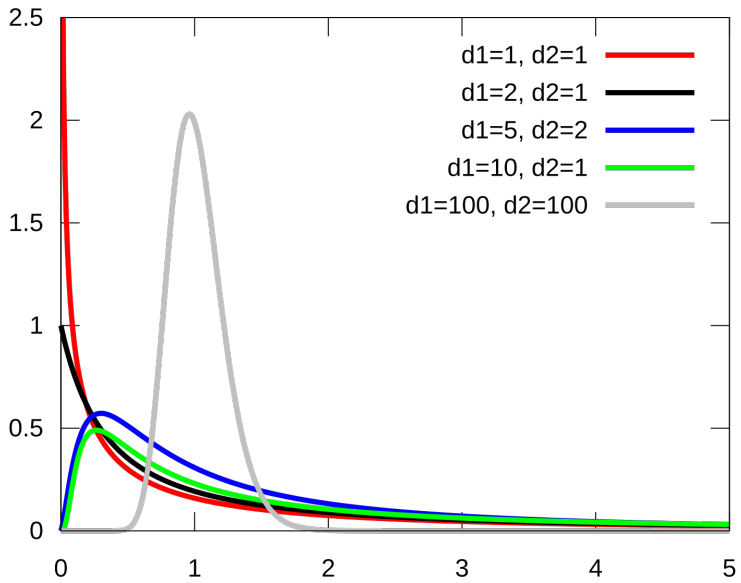
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- Mean: 0 for $\nu > 1$; undefined otherwise.
- Variance: $\frac{\nu}{\nu-2}$ for $\nu > 2$; ∞ for $\nu \in (1,2]$; undefined otherwise.
- Limits: $\nu \to 1$: Cauchy distribution; $\nu \to \infty$: $\mathcal{N}(0,1)$.

Celebrities+: F distribution ($d = 1$)



- Def ($d_1, d_2 \in \mathbb{Z}^+$ degrees of freedom):

$$X = \frac{U_1/d_1}{U_2/d_2},$$

where $U_1 \sim \chi^2_{d_1}$ and $U_2 \sim \chi^2_{d_2}$ independent.

Celebrities+: F distribution $(d = 1)$ – continued

- Pdf:

$$f_{d_1,d_2}(x) = \frac{\sqrt{\frac{(d_1x)^{d_1}(d_2)^{d_2}}{(d_1x+d_2)^{d_1+d_2}}}}{xB\left(\frac{d_1}{2}, \frac{d_2}{2}\right)}, \quad x \in (0, \infty) \text{ for } d_1 = 1, \text{ else } x \in [0, \infty).$$

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- Mean and variance:

$$\mathbb{E}[X] = \frac{d_2}{d_2 - 2} \text{ for } d_2 > 2, \quad \text{var}(X) = \frac{2d_2^2(d_1 + d_2 - 2)}{d_1(d_2 - 2)^2(d_2 - 4)} \text{ for } d_2 > 4.$$

Linear regression model

- Model:

$$Y = f(X) + \epsilon,$$
$$f(X) = f_\beta(X) = \beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p.$$

with $Y \in \mathbb{R}$, $X = (X_i)_{i=1}^p \in \mathbb{R}^p$, $\beta = (\beta_i)_{i=0}^p \in \mathbb{R}^{p+1}$, $\epsilon \sim \mathcal{N}(0, \sigma^2)$.

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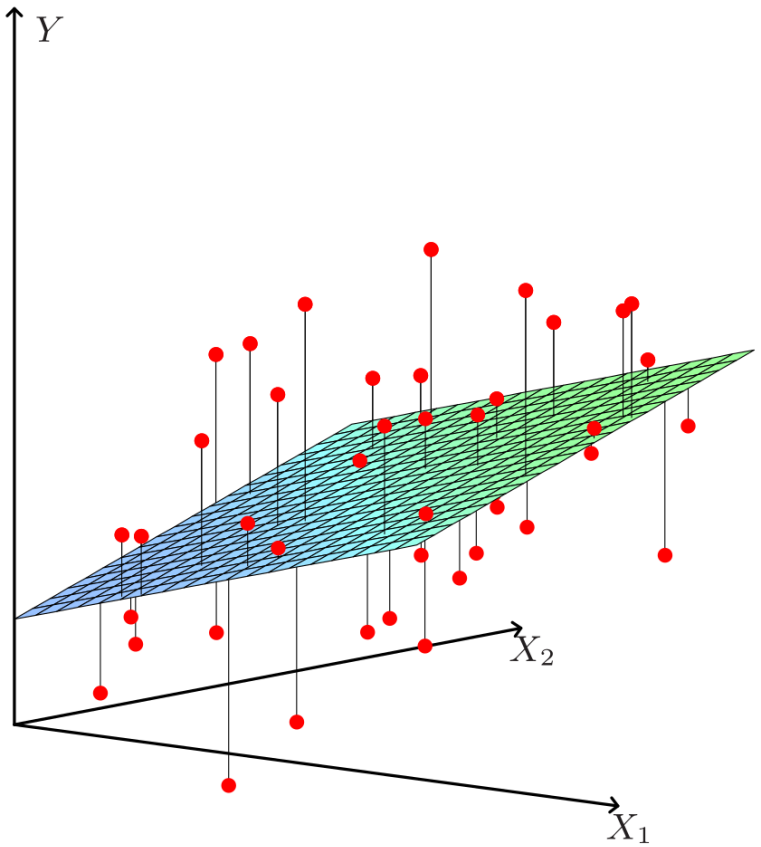
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- Simple and interpretable.
- Various generalizations (covered later).

Towards learning: objective function

- Given: $\{(\mathbf{x}_i, y_i)\}_{i=1}^n \subset \mathbb{R}^p \times \mathbb{R}$ i.i.d. realizations; $\mathbf{x}_i = (x_{i,j})_{j=1}^p$.
- Method: least squares, minimizing the residual sum of squares; demo ($p = 2$):



Objective function

Our goal:

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$$\begin{aligned}\mathrm{RSS}(\beta) &= \sum_{i\in[n]} [y_i - f_\beta(\mathbf{x}_i)]^2 = \sum_{i\in[n]} \left[y_i - \beta_0 - \sum_{j\in[p]} x_{i,j}\beta_j \right]^2 \\ &= \|\mathbf{y} - \mathbf{X}\beta\|_2^2 \to \min_{\beta\in\mathbb{R}^{p+1}},\end{aligned}$$

where $\mathbf{y} = (y_i)_{i=1}^n \in \mathbb{R}^n$, $\mathbf{X} = \left(\tilde{\mathbf{x}}_i^\top\right)_{i=1}^n \in \mathbb{R}^{n\times(p+1)}$, $\tilde{\mathbf{x}}_i = \begin{pmatrix} 1 \\ \mathbf{x}_i \end{pmatrix} \in \mathbb{R}^{p+1}$.

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Note: residual standard error (normalized RSS):

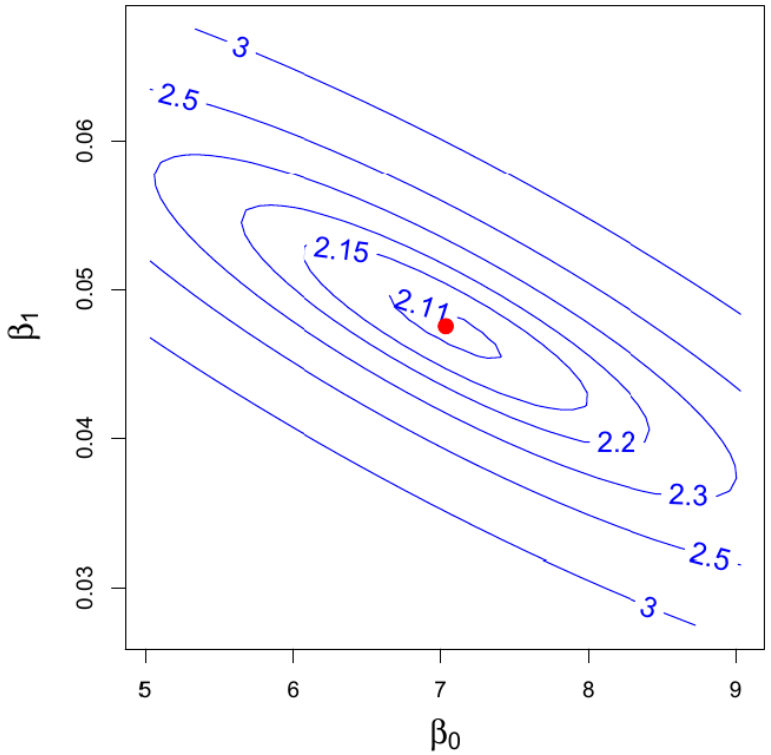
$$\mathrm{RSE} = \sqrt{\frac{1}{n-p-1}\mathrm{RSS}}.$$

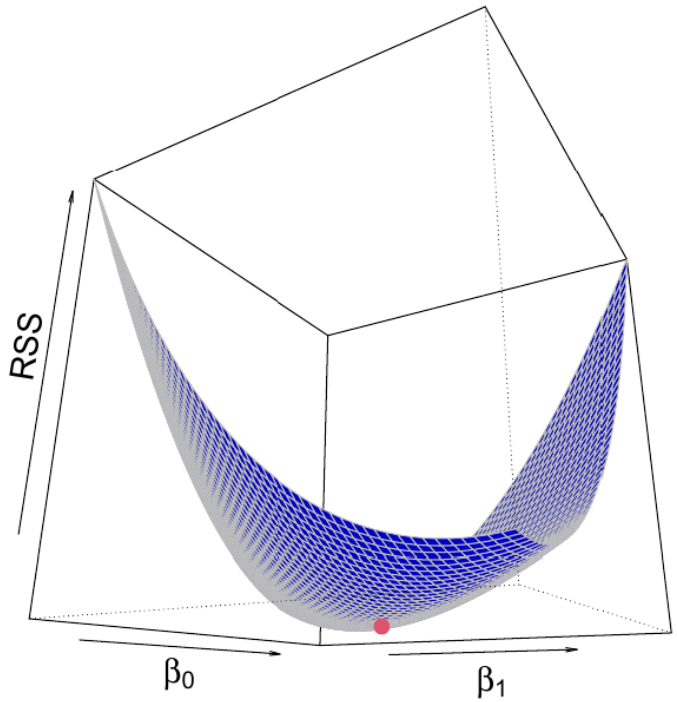
Optimization: visual illustration

With the aim

$$\text{sales} \approx \beta_0 + \beta_1 \times \text{TV},$$

the objective





Questions

In the example of

$$\text{sales} \approx \beta_0 + \beta_1 \times \text{TV} + \beta_2 \times \text{radio} + \beta_3 \times \text{newspaper},$$

natural questions:

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5. Would non-linearity help? $\Rightarrow$ polynomial regression, feature map.

Solution

$$\min_{\beta \in \mathbb{R}^{p+1}} \mathrm{RSS}(\beta) := \|\mathbf{y} - \mathbf{X}\beta\|_2^2$$

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Solution – continued

- Positive semi-definiteness: $\mathbf{X}^\top \mathbf{X} \succcurlyeq \mathbf{0}$. Indeed, for any $\mathbf{v} \in \mathbb{R}^{p+1}$

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- Normal equations [after taking $(\cdot)^\top$]:

$$-2\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\beta) = \mathbf{0} \overset{/-2}{\Longleftrightarrow} \mathbf{X}^\top (\mathbf{y} - \mathbf{X}\beta) = \mathbf{0}.$$

  Intuition (recall cos), columns of $\mathbf{X} \perp \mathbf{y} - \mathbf{X}\beta$.

Solution – continued

From the normal equations:

$$\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\beta) = \mathbf{0} \Leftrightarrow \mathbf{X}^\top\mathbf{X}\beta = \mathbf{X}^\top\mathbf{y}.$$

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$$\hat{\mathbf{y}} = \mathbf{X}\hat{\beta} = \underbrace{\mathbf{X}\left(\mathbf{X}^\top\mathbf{X}\right)^{-1}\mathbf{X}^\top}_{=:\mathbf{H}}\mathbf{y}.$$

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- Prediction on a new point $\mathbf{x}_0$: $\hat{f}(\mathbf{x}_0) = (\tilde{\mathbf{x}}_0)^\top\hat{\beta}$.

Solution: simpler form for $p = 1$

In case of $p = 1$:

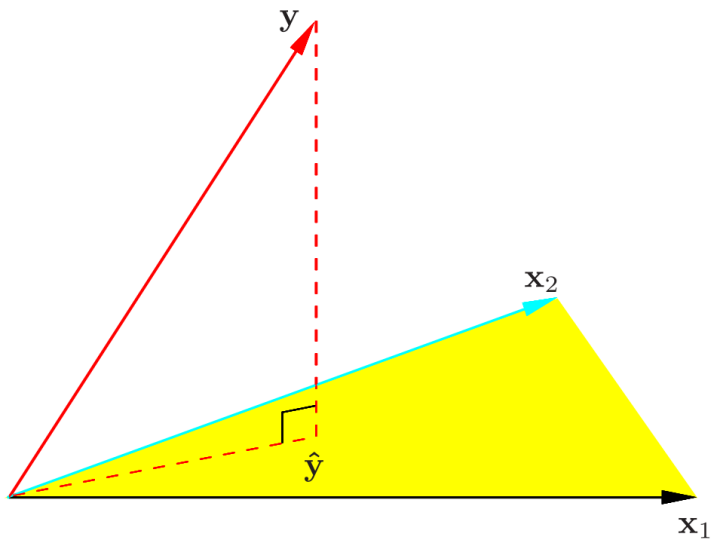
$$\bar{x} = \frac{1}{n} \sum_{i \in [n]} x_i, \qquad \bar{y} = \frac{1}{n} \sum_{i \in [n]} y_i,$$

the coefficients $\hat{\beta} = \left(\hat{\beta}_i\right)_{i=0}^{1} \in \mathbb{R}^2$ can be computed as

$$\hat{\beta}_1 = \frac{\sum_{i \in [n]} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i \in [n]} (x_i - \bar{x})^2},$$
$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Solution: visually

Orthogonality: $\mathbf{X}^\top \left(\mathbf{y} - \underbrace{\mathbf{X}\hat{\beta}}_{\hat{\mathbf{y}}} \right) = \mathbf{0}$.



H is also called projection matrix.

How to recognize an orthogonal projection?

$\mathbf{P}^2 = \mathbf{P} = \mathbf{P}^\top$.

The $\exists \left(\mathbf{X}^\top \mathbf{X}\right)^{-1}$ assumption

If $\nexists \left(\mathbf{X}^\top \mathbf{X}\right)^{-1}$:

- $\hat{\boldsymbol{\beta}}$ (coefficients): might not be unique.
- $\hat{\mathbf{y}}$ (projected $\mathbf{y}$): is still unique.
- In practice, typically regression softwares notice & handle it.



Recall the model, fixed design

- Model (with $\mathbf{X}$ fixed), and coefficient estimate

$$\mathbf{y} = \mathbf{X}\beta + \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I}); \qquad \hat{\beta} = \left(\mathbf{X}^\top\mathbf{X}\right)^{-1}\mathbf{X}^\top\mathbf{y}.$$

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Distributional properties

Variance:

- estimate

$$\hat{\sigma}^2 = \frac{1}{n-p-1} \sum_{i \in [n]} (y_i - \hat{y}_i)^2.$$

Note: factor $\frac{1}{n-p-1}$ makes sure that $\mathbb{E}\left[\hat{\sigma}^2\right] = \sigma^2$ (unbiased).

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Combining this, with $\hat{\boldsymbol{\beta}} \sim \mathcal{N}\left(\boldsymbol{\beta}, \left(\mathbf{X}\mathbf{X}^\top\right)^{-1} \sigma^2\right)$, and using the independence of $\hat{\sigma}^2$ and $\hat{\boldsymbol{\beta}}$, we get for $S \subseteq [p]$

1. test for $\boldsymbol{\beta}_S = \mathbf{0}$,
2. confidence interval for $\boldsymbol{\beta}_S$.

Test $\beta_j = 0$, $j \in [p]$ fixed, i.e. $S := \{j\}$

- Aim:
  - H_0 (null hypothesis): X_j has no effect on Y, i.e.

$$\beta_j = 0.$$

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$$z_j = \frac{\hat{\beta}_j}{\hat{\sigma}\sqrt{v_j}}, \qquad v_j = \mathbf{e}_j^\top \operatorname{diag}\left(\left(\mathbf{X}^\top \mathbf{X}\right)^{-1}\right).$$

Nickname: $\mathrm{SE}(\hat{\beta}_j) = \hat{\sigma}\sqrt{v_j}$ (**standard error** associated with $\hat{\beta}_j$).

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Under H_0: $z_j \sim t_{n-p-1}$.

Test $\beta_j = 0$: given level $\alpha \in (0, 1)$, say $\alpha = 0.01, 0.05$

- Idea: we reject H_0 if $|z_j|$ is 'big'.
- In terms
  - quantile: Determine (read out from sheet) the quantile s.t.

$$P(|t_{n-p-1}| \leq q) = 1 - \alpha.$$

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  - p-value:

$$\text{p-value} := P(|t_{n-p-1}| \geq z_j)$$

    Decision: if p-value $< \alpha$ (very unlikely observation), then reject H_0.

Standard errors for $p = 1$

They take a simple form:

$$\mathrm{SE}^2\left(\hat{\beta}_0\right) = \hat{\sigma}^2\left[\frac{1}{n} + \frac{\bar{x}}{\sum_{i\in[n]}(x_i - \bar{x})^2}\right], \quad \mathrm{SE}^2\left(\hat{\beta}_1\right) = \frac{\hat{\sigma}^2}{\sum_{i\in[n]}(x_i - \bar{x})^2}.$$

Intuition:

- As x_i is more spread out, the denominator of $\mathrm{SE}^2\left(\hat{\beta}_1\right)$ increases $\Rightarrow \mathrm{SE}^2\left(\hat{\beta}_1\right)$ decreases.

Test $\beta_S = \mathbf{0}$; $S \subseteq [p]$ fixed; given level $\alpha \in (0, 1)$

- We have a
  - model with $p_1 + 1$ parameters.
  - smaller nested model $p_0 + 1$ parameters.

  i.e. $p_1 - p_0$ parameters constrained to be zero (H_0; aim: elimination).
- F-statistic:

$$F = \frac{(\mathrm{RSS}_0 - \mathrm{RSS}_1)/(p_1 - p_0)}{\mathrm{RSS}_1/(n - p_1 - 1)}$$

$\mathrm{RSS}_i := \mathrm{RSS}$ for the model with $p_i + 1$ parameters ($i \in \{0, 1\}$).

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Under H_0 (smaller model is correct): $F \sim F_{p_1 - p_0, n - p_1 - 1}$.

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- Notes:
  1. using $z_j \Leftrightarrow$ F-statistic for dropping one coefficient β_j.
  2. for large n, $F_{p_1 - p_0, n - p_1 - 1} \approx \chi^2_{p_1 - p_0}/(p_1 - p_0)$.

Until now: $\boldsymbol{\beta} \sim \mathcal{N} \Rightarrow$ test $\boldsymbol{\beta}_S = \mathbf{0}$

- Recall:

$$\hat{\boldsymbol{\beta}} \sim \mathcal{N}\left(\boldsymbol{\beta}, \left(\mathbf{X}\mathbf{X}^\top\right)^{-1} \sigma^2\right).$$

- $\Rightarrow$ Now: confidence interval/set.
  1. single coordinate: $1-\alpha$ confidence interval for β_j

$$\left(\hat{\beta}_j - z^{(1-\frac{\alpha}{2})} \mathrm{SE}\left(\hat{\beta}_j\right), \hat{\beta}_j + z^{(1-\frac{\alpha}{2})} \mathrm{SE}\left(\hat{\beta}_j\right)\right)$$

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$$\alpha = 0.05, \qquad z^{(1-0.025)} = 1.96 \approx 2,$$
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$\Rightarrow$ For an $\approx 95\%$ confidence interval, standard practice:

$$\left(\hat{\beta}_j - 2\mathrm{SE}\left(\hat{\beta}_j\right), \hat{\beta}_j + 2\mathrm{SE}\left(\hat{\beta}_j\right)\right).$$

Confidence set: whole β

Recall:

$$\hat{\beta} \sim \mathcal{N}\left(\beta, \left(\mathbf{X}\mathbf{X}^\top\right)^{-1}\sigma^2\right).$$

$\Rightarrow$ Confidence set for β:

$$C_\beta = \left\{\beta \in \mathbb{R}^{p+1} \,:\, \left(\beta - \hat{\beta}\right)^\top \mathbf{X}^\top\mathbf{X}\left(\beta - \hat{\beta}\right) \leq \hat{\sigma}^2\left(\chi^2_{p+1}\right)^{(1-\alpha)}\right\},$$

where $\left(\chi^2_k\right)^{(1-\alpha)}$ is the $1-\alpha$ quantile of the distribution χ^2_k.

Confidence set: whole $\boldsymbol{\beta}$

Recall:

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$\Rightarrow$ Confidence set for $\boldsymbol{\beta}$:

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What kind of set is $C_{\boldsymbol{\beta}}$?

Confidence set: whole β – continued

- Obtained confidence set:

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- Classically, $\mathbf{M} = \mathbf{I}$.

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$\Rightarrow$ C_β is a ball with center $\hat{\beta}$ (or in the classical sense an ellipsoid)

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Once we checked $\beta \neq \mathbf{0}$

2 widely-used measures of fit:

1. RSE (recall):

$$\mathrm{RSE} = \sqrt{\frac{1}{n-p-1}\mathrm{RSS}}, \qquad \mathrm{RSS} = \sum_{i \in [n]} (y_i - \hat{y}_i)^2.$$

It measures the lack of fit to data in the units of Y.

Once we checked $\beta \neq \mathbf{0}$ – continued

2. R^2:

$$\mathrm{R}^2 = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}}, \qquad \mathrm{RSS} = \sum_{i \in [n]} (y_i - \hat{y}_i)^2, \qquad \mathrm{TSS} = \sum_{i \in [n]} (y_i - \bar{y})^2.$$

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- RSS: unexplained variability in the response left after regression,
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- Value: $\mathrm{R}^2 \in [0, 1]$. Relation to empirical correlation ($p = 1$):

$$r = \widehat{\mathrm{corr}}(X, Y) = \frac{\sum_{i \in [n]} (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i \in [n]} (x_i - \bar{x})^2 \sum_{i \in [n]} (y_i - \bar{y})^2}}.$$

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With this notation, $\mathrm{R}^2 = r^2$.

Variants: qualitative predictors

- Aim:

  credit card balance $\approx \beta_0 + \beta_1 \times$ being the owner of a house.

- Dummy variables:

$$x_i = \begin{cases} 1 & \text{if the } i^{th} \text{ person owns a house} \\ 0 & \text{otherwise} \end{cases}$$

- Resulting model/observations:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad i \in [n].$$

Variants: interaction/synergy terms

- Standard model:

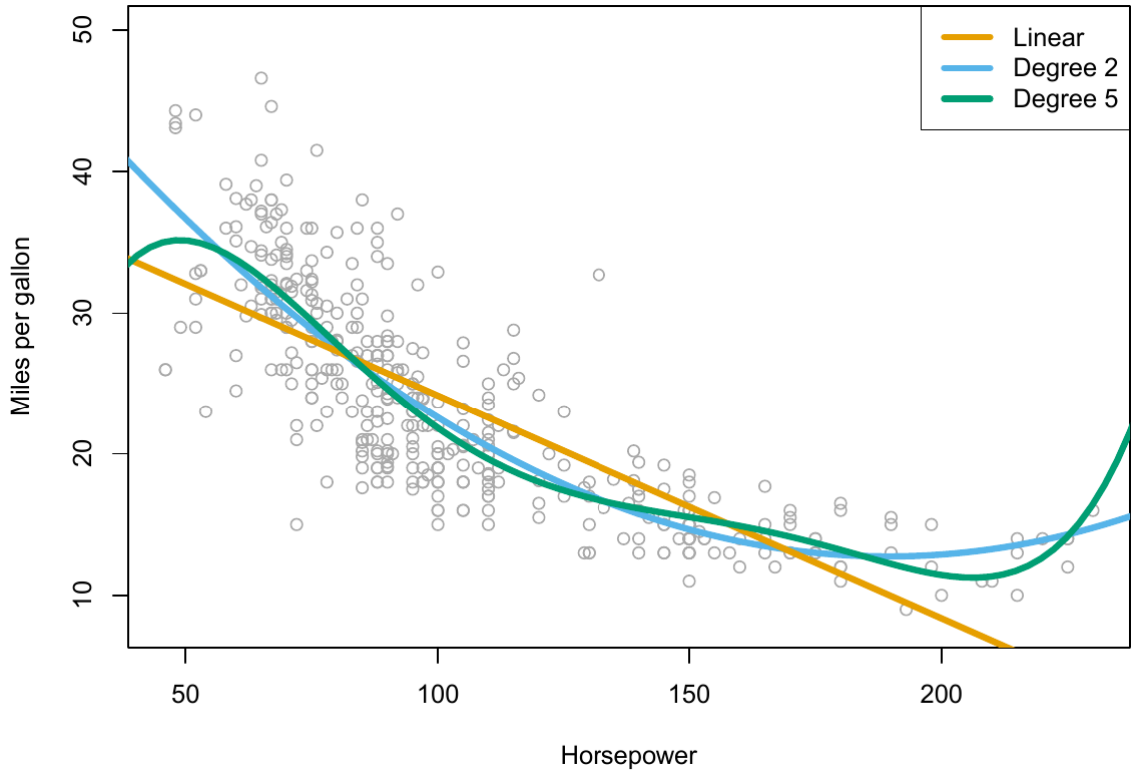
$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon.$$

- Model with interaction between X_1 and X_2:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \epsilon.$$

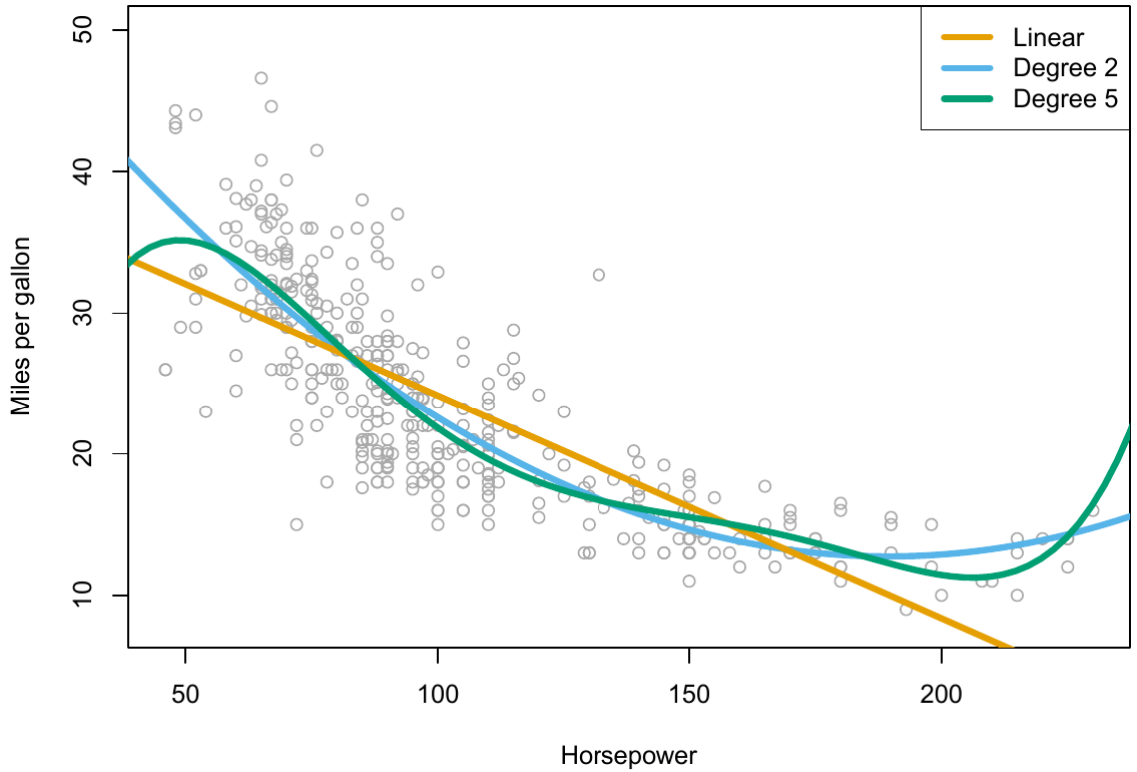
Variants: polynomial features

- Example: mpg (gas mileage in miles per gallon) versus horsepower.



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Instead of a linear fit (which looks off):

$$\text{mpg} = \beta_0 + \beta_1 \times \text{horsepower} + \beta_2 \times \text{horsepower}^2 = \langle \boldsymbol{\beta}, \varphi(X) \rangle,$$

$$\varphi(X) = \left(1, X, X^2\right).$$

Linear regression vs kNN regression

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  - $\mathcal{N}_0 = K$ training observations closest to $\mathbf{x}_0$.

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Linear regression vs kNN regression

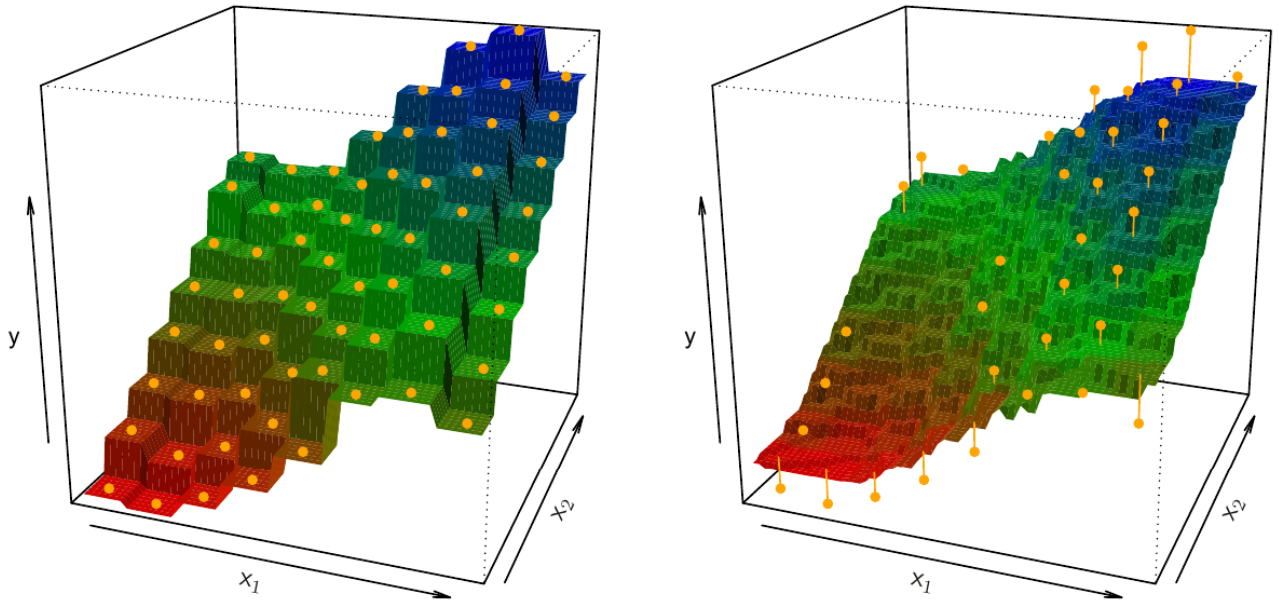
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It takes the average prediction of the KNNs.

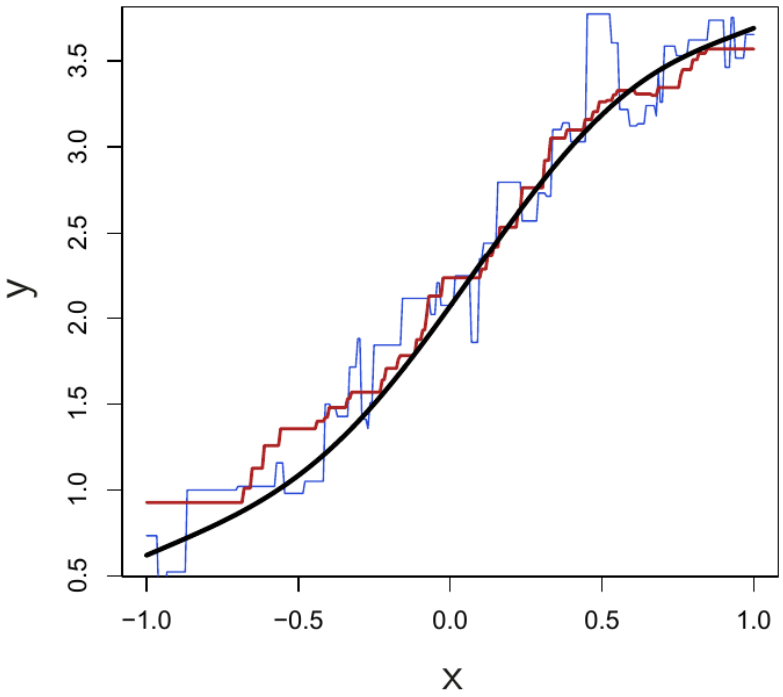
KNN regression: example with $p = 2$

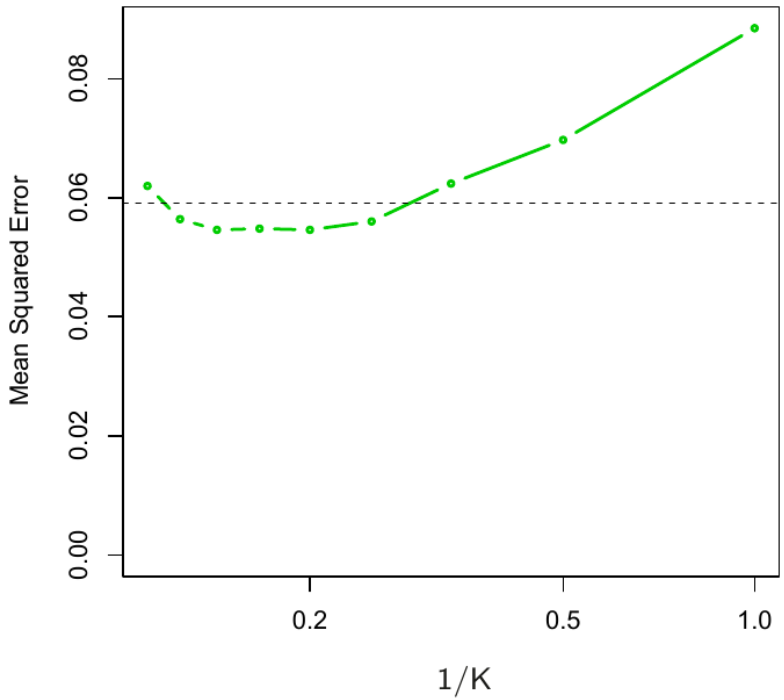
Left: $K = 1$ (rougher), right: $K = 9$ (smoother).



Comparison of linear and KNN regression-1

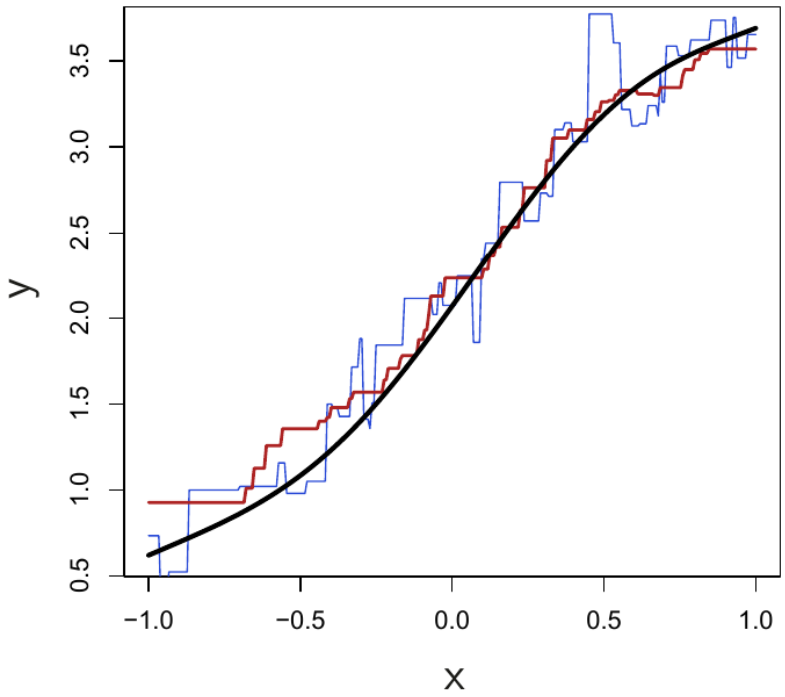
- setting: $\approx$ linear relation.
- left: fit with $K = 1$ (blue) and $K = 9$ (red); right: test MSE vs different K-s, with linear regression baseline (black).

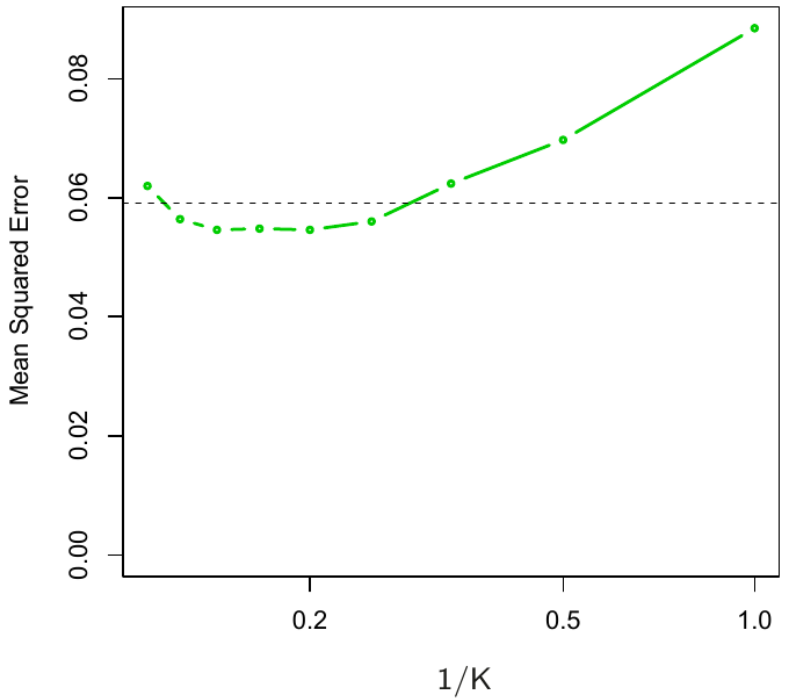




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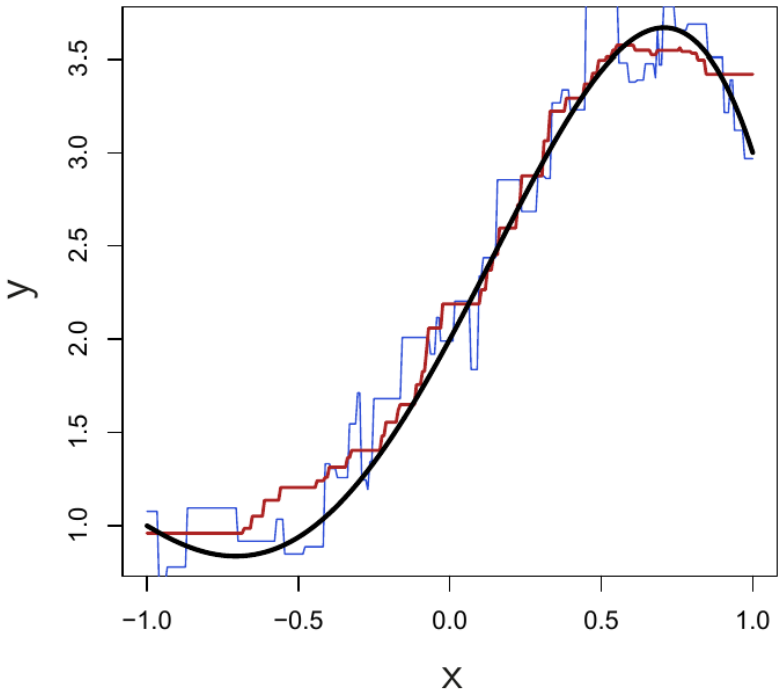


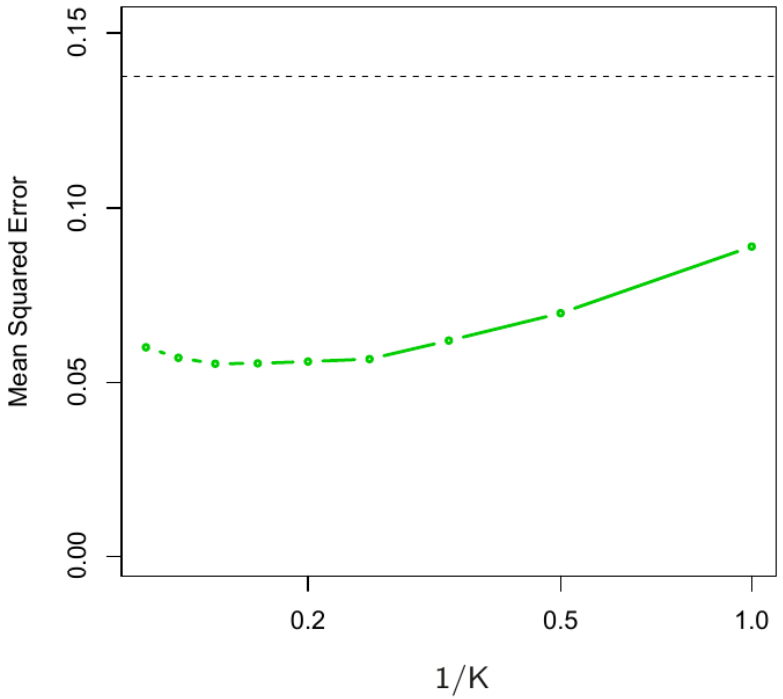


$\Rightarrow$ KNN regression: better for $K \geq 4$ (l.h.s. on the r.h.s.).

Comparison of linear and KNN regression-2

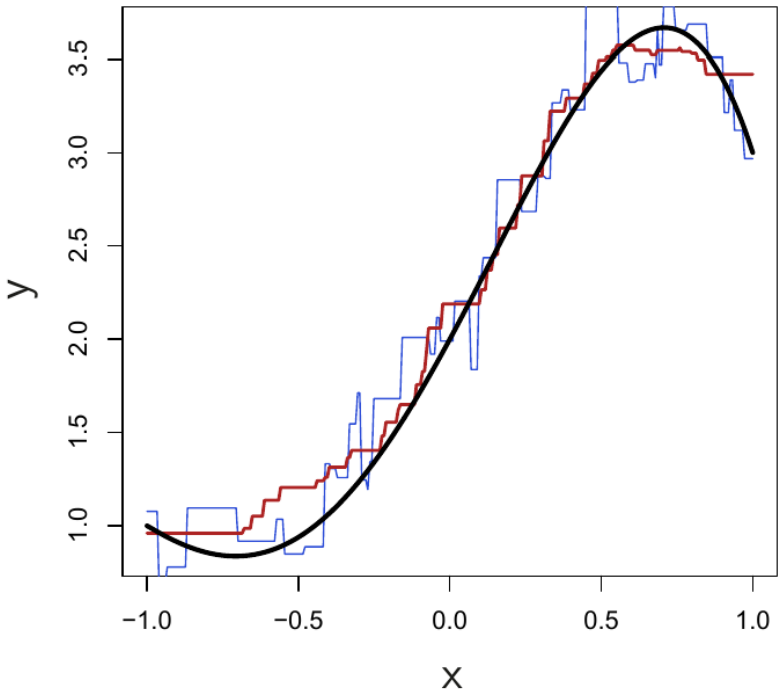
- setting: increased non-linearity.
- left: fit with $K = 1$ (blue) and $K = 9$ (red); right: test MSE vs different K-s, with linear regression baseline (black).

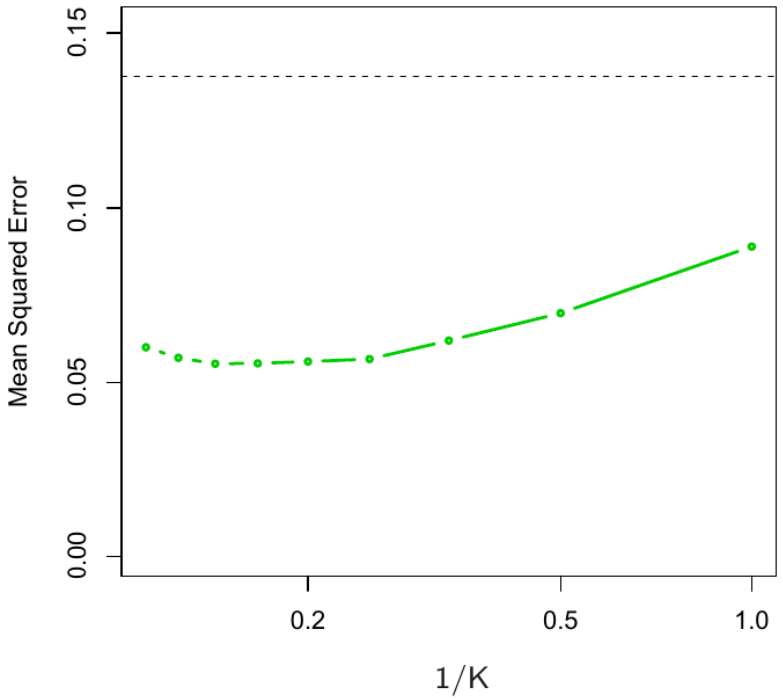




Comparison of linear and KNN regression-2

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- left: fit with $K = 1$ (blue) and $K = 9$ (red); right: test MSE vs different K-s, with linear regression baseline (black).





$\Rightarrow$ KNN regression: better overall; in higher dimension it can struggle (curse of dimensionality).

Local summary

- probability recap; key distributions: $\mathcal{N}$, χ^2, t, F,
- least squared problem: formulation, solution, projection view,
- testing, confidence intervals,
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